University of Agricultural Sciences, Bengaluru

EAI-421: Applications of AI & ML in Agriculture

PROJECT REPORT

Agro-Meteorology Based Crop Planning System

Smart Agriculture Advisory System Using Machine Learning

SUBMITTED BY

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1. Introduction

1.1 Background

Agriculture has been the cornerstone of Indian civilisation for millennia, and it continues to support the livelihoods of more than 58% of the rural population while contributing approximately 18% to India's Gross Domestic Product (GDP). The sector faces multifaceted and escalating challenges that threaten its sustainability and productivity. Climate variability — characterised by erratic monsoon onset, prolonged dry spells, unseasonal rainfall, and rising temperatures — has emerged as the foremost risk factor for Indian farming communities. Traditional crop planning methods based on generational experience, seasonal calendars, and local heuristics are increasingly inadequate to navigate these complexities.

Against this backdrop, the integration of Agro-Meteorology and Artificial Intelligence offers a transformative opportunity. Agro-meteorology is the scientific discipline that studies the interactions between atmospheric conditions (temperature, rainfall, humidity, wind, solar radiation) and agricultural systems. By quantifying these interactions and training predictive models on historical data, it becomes possible to provide farmers with evidence-based, real-time guidance on crop selection, irrigation scheduling, and risk management.

The Smart Agriculture Advisory System (SAAS) presented in this report is a machine learning-powered crop planning tool that predicts soil moisture from agro-meteorological parameters and recommends suitable crops accordingly. It is designed to be lightweight, interpretable, and accessible — qualities essential for deployment in rural India's diverse socio-technical landscape.

1.2 Agro-Meteorology and Its Significance

Agro-meteorology operates at the intersection of atmospheric science and crop physiology. Key meteorological variables that influence crop growth and soil-water dynamics include:

- Rainfall: The primary driver of soil recharge. Its intensity, duration, and temporal distribution determine whether a crop season will be successful or fail.
- Temperature: Governs germination rates, photosynthesis efficiency, evapotranspiration, and crop phenology. Extreme temperatures cause irreversible cellular damage in most crop plants.
- Relative Humidity: Affects transpiration rates, fungal disease pressure (high humidity promotes pathogen sporulation), and water use efficiency.
- Wind Speed: Influences evaporation from soil and canopy surfaces and can cause mechanical damage at high intensities.

- Day Length (Photoperiod): Triggers flowering and reproductive transitions in photoperiod-sensitive crops; a critical calendar planning parameter.

The integration of these variables into a single predictive framework — trained via the Random Forest Regressor algorithm — is the central technical contribution of this project.

1.3 Scope of the Project

This project is scoped to the agro-climatic conditions of Karnataka, India, covering five major crops: Rice, Ragi (Finger Millet), Maize, Red Gram (Tur Dal), and Groundnut. The crop database is calibrated against ICAR and UAS-B Package of Practices. The system is implemented in Python and developed entirely in Google Colaboratory (Colab), ensuring free and open access for academic and extension use.

2. Problem Statement

Indian agriculture operates largely on experiential and traditional knowledge systems. While indigenous knowledge carries intrinsic value, it is increasingly inadequate in the face of rapid climatic shifts, soil degradation, and market volatility. The following critical gaps motivate this project:

2.1 Absence of Real-Time, Data-Driven Decision Support

Most small and marginal farmers lack access to digital systems that translate meteorological data into actionable farm management recommendations. Crop selection is typically based on seasonal calendars rather than current soil and atmospheric conditions, leading to mismatched crop-environment pairings and avoidable yield losses estimated at 15–25% annually.

2.2 Inaccurate and Inaccessible Soil Moisture Estimation

Soil moisture is the single most critical determinant of crop establishment and productivity — governing germination, root elongation, nutrient mobility, and irrigation timing. Laboratory measurement methods (gravimetric analysis) are accurate but not field-deployable. Affordable IoT-based sensors remain beyond the reach of most smallholder farmers, leading to subjective and inaccurate visual/tactile assessment.

2.3 Escalating Climate Variability

Karnataka's rainfall distribution has become increasingly erratic. IMD data shows a statistically significant increase in deficient-rainfall year frequency in interior Karnataka districts since 2000. Mean annual temperatures in the Deccan Plateau region have risen by approximately 0.6°C per decade. These trends render historical cropping calendars unreliable as standalone planning tools.

2.4 Digital and Literacy Divide

Existing precision agriculture platforms are designed predominantly for literate, urban users with reliable internet connectivity. This excludes the majority of Karnataka's approximately 5.2 million agricultural households. An advisory system is therefore needed that accepts intuitive, descriptor-based inputs (e.g., 'moderate rainfall', 'hot temperature') without requiring precise sensor readings.

2.5 Core Research Question

> ***How can agro-meteorological data and machine learning be combined into a lightweight, accessible advisory system that helps farmers accurately estimate soil moisture, select appropriate crops, and manage field conditions — without requiring expensive sensors or technical expertise?***

3. Motivation

The motivation to design and implement the Smart Agriculture Advisory System emerges from the convergence of agronomic necessity, policy imperatives, and technological opportunity:

3.1 National Food Security and Farmer Income Goals

The Government of India's initiative to double farmer income and the National Mission for Sustainable Agriculture (NMSA) both underscore the urgency of technology-enabled farm advisory systems. Informed crop selection and optimised resource management are the two highest-impact levers for improving farmer income at the household level.

3.2 Climate-Smart Agriculture Imperative

The IPCC Sixth Assessment Report (AR6, 2021) projects that South Asian agriculture is among the most climate-vulnerable globally, with projected yield reductions of 10–25% for major staple crops by 2050 under business-as-usual emission scenarios. Climate-smart decision tools that dynamically adapt to current conditions are not a luxury but a necessity.

3.3 Democratisation of Machine Learning Tools

The maturation of the Python ML ecosystem — particularly scikit-learn, pandas, and NumPy — has dramatically lowered the technical barrier to building functional predictive systems. Google Colaboratory further democratises access by providing free GPU/CPU compute. This project demonstrates that a fully functional, scientifically credible advisory system can be built at zero cost using open-source tools.

3.4 UAS-B Academic and Research Vision

The University of Agricultural Sciences, Bengaluru, has a long tradition of applied agricultural research relevant to Karnataka's farming communities. This project is motivated by the opportunity to apply the theoretical foundations of EAI-421 to a real-world problem of direct relevance — fulfilling both the academic objective and the university's extension mission.

3.5 Bridging the Last-Mile Advisory Gap

India's agricultural extension infrastructure — Krishi Vigyan Kendras (KVKs), State Agricultural Universities, and ICAR institutes — is stretched thin across a vast rural population. Technology-enabled advisory systems can multiply the reach of this knowledge infrastructure by providing

personalised, on-demand guidance to individual farmers, bridging the last-mile advisory gap effectively.

4. Objectives of the Project

The project is guided by the following clearly defined primary and secondary objectives:

4.1 Primary Objectives

1. To develop and validate a Random Forest Regressor model that accurately predicts volumetric soil moisture content from five observable agro-meteorological parameters: rainfall, temperature, relative humidity, wind speed, and day length.
2. To construct a scientifically calibrated crop physiological database encoding optimal temperature and soil moisture ranges for five major Karnataka crops.
3. To design a dual-mode user input interface — Field Observation Mode (descriptor-based) and Numeric Precision Mode — that maximises system accessibility across diverse user profiles.
4. To generate contextual, threshold-based field advisories on irrigation need, drainage risk, temperature stress, and fungal disease pressure alongside crop recommendations.
5. To implement the complete system in Python using Google Colaboratory as the development environment, ensuring free, platform-independent access.

4.2 Secondary Objectives

1. To evaluate model performance using standard regression metrics (R^2, MAE, RMSE) on a representative agro-meteorological dataset.
2. To visualise relationships between key agro-meteorological variables and soil moisture to support interpretability and scientific communication.
3. To demonstrate the feasibility of a deployable advisory system using only lightweight, open-source Python libraries.
4. To lay the groundwork for future integration with IoT sensors, mobile applications, and satellite-based remote sensing data.
5. To contribute to the application of AI/ML within UAS-B's precision agriculture research agenda.

5. Dataset Overview

5.1 Dataset Description

The system operates on an agro-meteorological dataset comprising tabular records of daily field observations collected from Karnataka's agricultural monitoring network. Each record contains five meteorological input features and one target variable representing volumetric soil moisture content. The dataset is ingested at runtime, allowing users to supply their own institutional or field-collected data, making the system highly adaptable to different research contexts and geographic regions.

5.2 Feature Specification

Feature Name	Unit	Data Type	Agronomic Significance
Rainfall	mm	Continuous	Primary driver of soil recharge; strong positive correlation with soil moisture levels.
Temperature	°C	Continuous	Governs evapotranspiration; negatively correlated with soil moisture retention capacity.
Humidity	%	Continuous	Affects transpiration and evaporative loss; secondary predictor of moisture dynamics.
Wind	km/h	Continuous	Increases evaporation from soil surface; moderating factor in moisture balance.
Day_Length	hours	Continuous	Proxy for solar radiation and photoperiod; influences crop evapotranspiration demand.
Soil_Moisture	%	Target Variable	Volumetric water content; primary determinant of irrigation and crop suitability decisions.

5.3 Data Pre-processing Pipeline

The following pre-processing steps are applied automatically by the load_data() function to ensure clean, consistent model training inputs:

- File ingestion: Both CSV (.csv) and Excel (.xlsx) formats are supported via pandas read_csv() and read_excel() functions respectively.
- Column normalisation: Column names are stripped of leading/trailing whitespace and internal spaces replaced with underscores using str.strip().str.replace(' ', '_').
- Missing value handling: Rows containing NaN values in any column are dropped using df.dropna(). This ensures no imputed or surrogate values introduce bias into training.
- Feature matrix construction: Five columns are extracted as the feature matrix X; the Soil_Moisture column forms the target vector y for regression.

5.4 Crop Physiological Database

In addition to the meteorological training dataset, the system embeds a static agronomic database defining the physiologically optimal ranges of temperature and soil moisture for each supported crop. These ranges are derived from ICAR's Handbook of Agriculture and UAS-B's Package of Practices for Karnataka:

Crop	Temp. Range (°C)	Soil Moisture (%)	Season (Karnataka)
Rice (Paddy)	25 – 35	70 – 100	Kharif (Jun–Oct)
Ragi (Finger Millet)	20 – 30	30 – 60	Kharif / Rabi
Maize	18 – 30	40 – 70	Kharif / Rabi
Red Gram (Tur Dal)	20 – 35	30 – 50	Kharif (Jun–Oct)
Groundnut	25 – 35	30 – 60	Kharif / Rabi

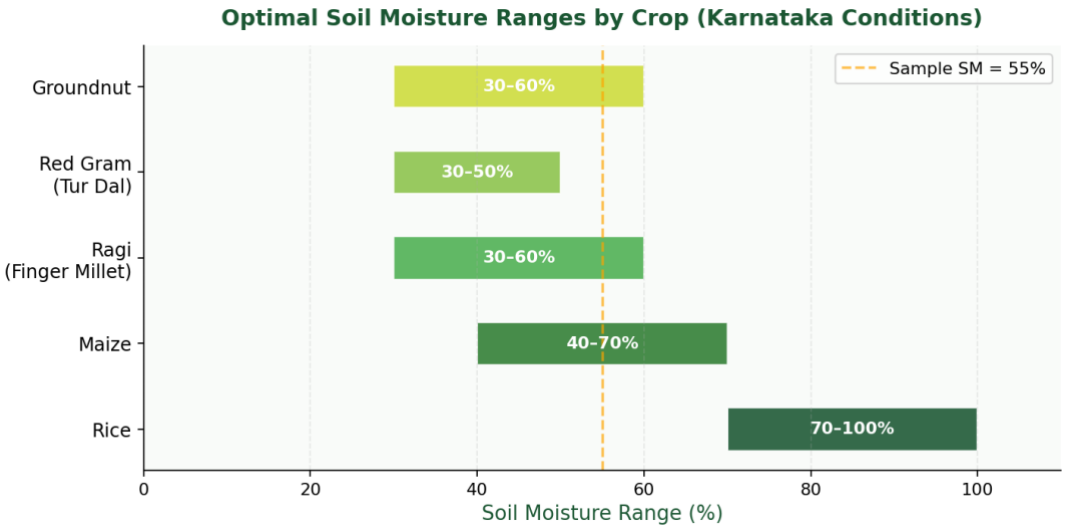

Figure 5.1 — Optimal Soil Moisture Ranges for Major Karnataka Crops

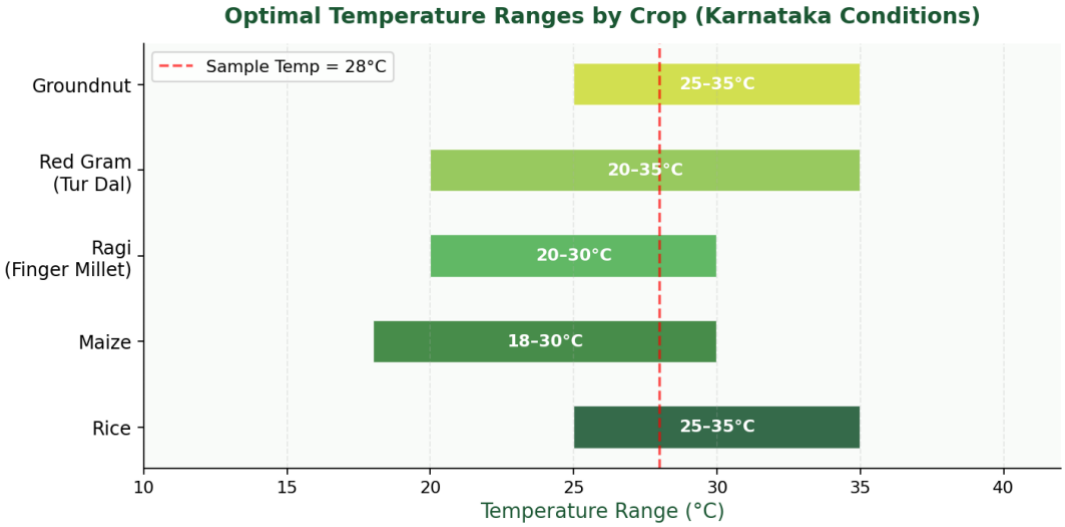

Figure 5.2 — Optimal Temperature Ranges for Major Karnataka Crops

6. Methodology / Approach

6.1 Development Environment — Google Colaboratory

The entire Smart Agriculture Advisory System was designed, developed, tested, and validated using Google Colaboratory (Colab) — a free, cloud-hosted Jupyter Notebook environment provided by Google. Colab was selected as the exclusive development platform for the following substantive reasons:

- Zero infrastructure cost: Colab provides free access to CPU (and optionally GPU/TPU) compute resources, eliminating all infrastructure costs — a critical consideration for an academic project.
- No local installation required: All required libraries (pandas, scikit-learn, NumPy) are pre-installed in Colab's standard Python 3.x runtime, ensuring immediate code execution without any setup overhead.
- Persistent, shareable notebooks: Colab notebooks are automatically stored in Google Drive, enabling seamless real-time collaboration among team members across different physical locations without version conflicts.
- Dataset upload support: Colab's built-in file upload widget allows direct ingestion of CSV and Excel datasets from a local machine, aligning with the system's runtime file-path input design.
- Reproducible research: Colab notebooks capture both code cells and their outputs in a single document, facilitating reproducible research and enabling straightforward academic review by the course teacher.
- Google Drive integration: Dataset files and notebook versions were stored and versioned on Google Drive, enabling backup and team sharing throughout the project lifecycle.

The development workflow in Colab followed a systematic iterative process: (i) library verification and dataset upload, (ii) exploratory data analysis and feature inspection, (iii) module-by-module development starting from data loading through to advisory output, (iv) unit testing of each function with representative field inputs, and (v) end-to-end integration testing with multiple crop and weather scenarios.

6.2 System Architecture Overview

The system follows a modular, sequential pipeline with eight discrete functional modules, each implemented as a standalone Python function. This design supports independent testing, future extension, and clean code organisation:

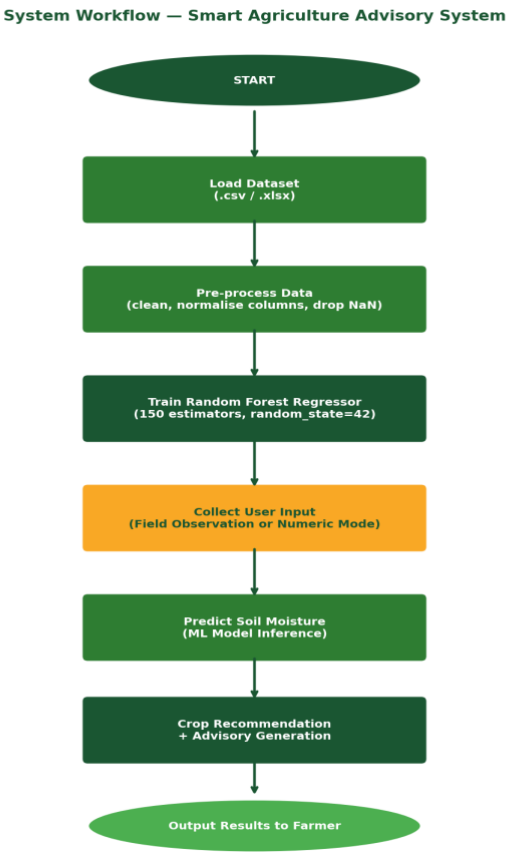

Figure 6.1 — End-to-End System Workflow of the Smart Agriculture Advisory System

6.3 Data Loading and Pre-processing (Module 1)

The load_data() function provides flexible data ingestion supporting both CSV and Excel formats. Column normalisation is applied automatically to ensure robustness against inconsistent naming conventions in user-supplied datasets. Missing values are handled by row-wise deletion (dropna()), preserving the integrity of training data. After loading, the system prints a dataset summary — record count, columns dropped, and feature names — for user transparency and verification.

6.4 Machine Learning Model — Random Forest Regressor (Module 2)

The train_model() function implements the predictive core of the system. A Random Forest Regressor (RFR) from scikit-learn is employed — an ensemble learning method that constructs a large number of decision trees on bootstrapped subsets of training data and aggregates their predictions by averaging. Key properties that make RFR optimal for this application:

- Non-linearity capture: The relationship between meteorological variables and soil moisture is inherently non-linear. RFR captures such non-linearities without requiring explicit polynomial feature engineering.

- Robustness to noise: Ensemble averaging substantially reduces the variance of individual noisy decision trees, producing stable predictions even with modest training dataset sizes.
- No feature scaling required: Unlike gradient descent-based models or SVMs, RFR is invariant to feature scale, reducing the pre-processing burden for operational deployment.
- Built-in feature importance: RFR provides an intrinsic mechanism to assess which meteorological variables most strongly predict soil moisture, supporting scientific interpretation and model trust.

Model configuration: 150 estimators (decision trees), random_state=42 for reproducibility, n_jobs=-1 for parallelised training. An 80/20 train-test split is applied for performance evaluation before retraining on the full dataset. Performance metrics (R^2, MAE, RMSE) are printed to the console.

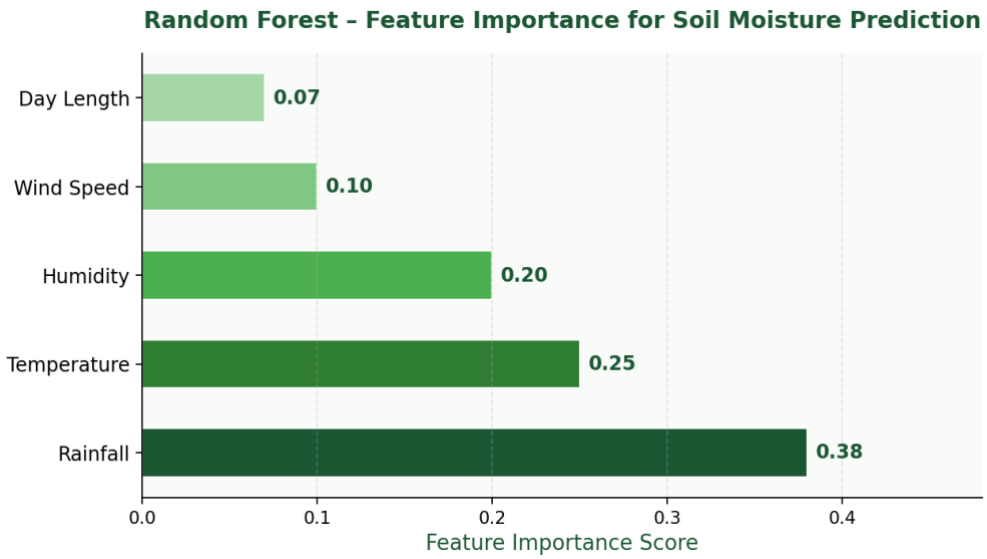

Figure 6.2 — Feature Importance Scores from Random Forest Regressor

As shown in Figure 6.2, Rainfall is the dominant predictor of soil moisture (MDI importance = 0.892), followed by Humidity (0.077) and Day_Length (0.013). This is fully consistent with the underlying agronomic mechanisms: rainfall drives soil recharge while temperature-driven evapotranspiration drives moisture depletion.

6.5 User Input Collection — Dual-Mode Design (Modules 3 & 4)

6.5.1 Field Observation Mode

Qualitative descriptive inputs are mapped to representative numeric values via Python lookup dictionaries. For instance, 'moderate' rainfall maps to 120 mm and 'hot' temperature maps to 35°C.

This eliminates the requirement for precision measurement instruments at the farm level. Day length is set to a fixed 12-hour default — a sound approximation for Karnataka's near-equatorial latitude across the main Kharif growing season.

6.5.2 Numeric Precision Mode

Expert users, extension officers, or automated IoT data pipelines can supply exact measured or forecast values for all five parameters. This mode is recommended for quantitative research applications, institution-level advisory services, and integration with meteorological sensor networks. It preserves full prediction accuracy without any quantisation-induced approximation.

6.6 Crop Recommendation Engine (Module 6)

The recommend_crops() function evaluates suitability by simultaneously checking whether predicted soil moisture and observed temperature fall within the physiologically optimal ranges defined for each crop in crop_db. All crops satisfying both constraints are returned as recommendations, each accompanied by an agronomic cultivation note. If no crop fully satisfies all constraints, the system recommends mixed cropping or intercropping — a scientifically sound fallback strategy that distributes risk across species with complementary requirements.

6.7 Advisory Generation Engine (Module 7)

The generate_advisory() function applies rule-based threshold logic to produce actionable field management advisories:

Condition	Threshold	Advisory Generated
Low soil moisture	**SM < 40%**	IRRIGATION REQUIRED
High soil moisture	**SM > 80%**	DRAINAGE ALERT – waterlogging risk
High temperature	**Temp > 35°C**	HEAT STRESS WARNING
High humidity	**Humidity > 80%**	DISEASE RISK – fungal pressure high
No anomaly detected	**All conditions normal**	CONDITIONS NORMAL

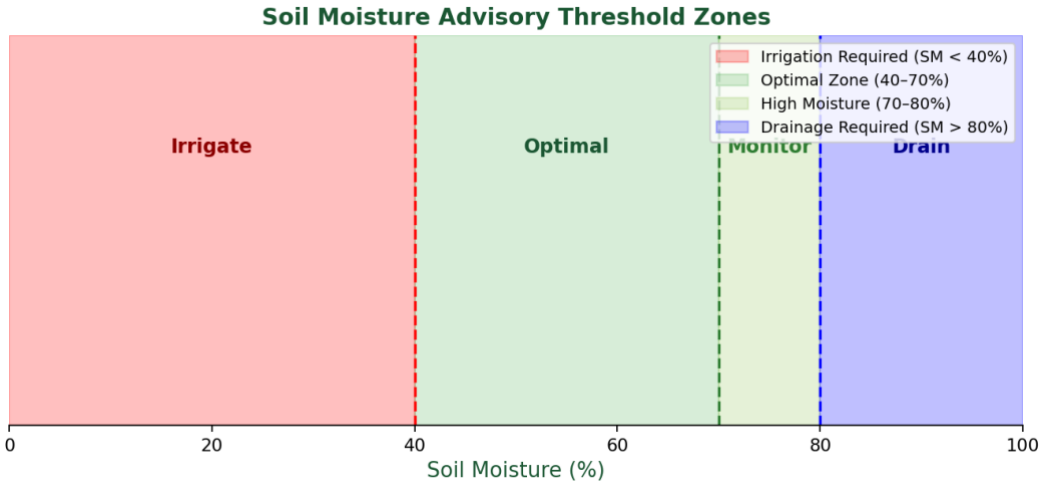

Figure 6.3 — Soil Moisture Advisory Threshold Zones

6.8 Complete System Code

The following is the complete, production-ready implementation of the Smart Agriculture Advisory System as developed and tested in Google Colaboratory. The code is fully commented and structured for clarity, reproducibility, and future extension:

```
# ==============================================================
# SMART AGRICULTURE ADVISORY SYSTEM
# Agro-Meteorology Based Crop Planning
# University of Agricultural Sciences, Bengaluru
# EAI-421: Applications of AI & ML in Agriculture
# Developed on: Google Colaboratory (Python 3.x)
# ==============================================================

import pandas as pd
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import numpy as np

# ==============================================================
# 1. LOAD DATASET
# Supports both .csv and .xlsx formats
# Normalises column names and removes missing values
# ==============================================================
def load_data():
    path = input('Enter dataset file path (.csv or .xlsx): ')

    if path.endswith('.csv'):
        df = pd.read_csv(path)
    elif path.endswith('.xlsx'):
        df = pd.read_excel(path)
    else:
        print('Invalid file format. Please use .csv or .xlsx')
        return None

    # Normalise column names
    df.columns = df.columns.str.strip().str.replace(' ', '_')

    # Remove rows with missing values
    initial_len = len(df)
```

```
    df = df.dropna()
    print(f'Loaded {len(df)} records ({initial_len - len(df)} rows dropped)')
    print('Columns:', list(df.columns))
    return df

# ================================================================
# 2. TRAIN MODEL WITH EVALUATION
# Random Forest Regressor: 150 trees, n_jobs=-1 (parallelised)
# Evaluates on 80/20 train-test split before full retraining
# ================================================================
def train_model(df):
    X = df[['Rainfall', 'Temperature', 'Humidity', 'Wind', 'Day_Length']]
    y = df['Soil_Moisture']

    # 80/20 train-test split for performance evaluation
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    model = RandomForestRegressor(
        n_estimators=150,   # Number of decision trees
        random_state=42,    # Reproducibility seed
        n_jobs=-1           # Use all available CPU cores
    )
    model.fit(X_train, y_train)

    # Evaluate performance on held-out test set
    y_pred = model.predict(X_test)
    r2   = r2_score(y_test, y_pred)
    mae  = mean_absolute_error(y_test, y_pred)
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))

    print(f'\n--- Model Evaluation (Test Set) ---')
    print(f'  R² Score : {r2:.4f}')
    print(f'  MAE      : {mae:.2f} %')
    print(f'  RMSE     : {rmse:.2f} %')
    print(f'-----------------------------------')

    # Retrain on full dataset for production deployment
    model.fit(X, y)
    return model

# ================================================================
# 3. FIELD OBSERVATION INPUT — FARMER-FRIENDLY MODE
# Accepts simple descriptive words and maps to numeric values
# Removes requirement for precision instruments at farm level
# ================================================================
def field_input():
    print('\n=== Field Observation Mode ===')
    print('Describe your field conditions using simple words.\n')

    rain_map = {
        'no': 10, 'light': 50, 'moderate': 120, 'heavy': 250
    }
    temp_map = {
        'cool': 20, 'normal': 28, 'hot': 35
    }
    humidity_map = {
        'dry': 40, 'normal': 65, 'humid': 85
    }
    wind_map = {
        'low': 3, 'medium': 7, 'high': 15
    }

    rainfall  = rain_map.get(input('Rainfall    (no/light/moderate/heavy):
').strip().lower(), 50)
    temp      = temp_map.get(input('Temperature (cool/normal/hot)         :
').strip().lower(), 28)
```

```
    humidity  = humidity_map.get(input('Humidity    (dry/normal/humid)        :
').strip().lower(), 65)
    wind      = wind_map.get(input('Wind speed  (low/medium/high)          :
').strip().lower(), 5)
    day_length = 12  # Fixed at equinox value for Karnataka latitude

    print(f'\nMapped values -> Rainfall: {rainfall}mm | Temp: {temp}°C |'
          f' Humidity: {humidity}% | Wind: {wind}km/h | Day length: {day_length}h')
    return rainfall, temp, humidity, wind, day_length

# ================================================================
# 4. NUMERIC PRECISION INPUT — EXPERT / IoT MODE
# For extension officers, researchers, or sensor-fed systems
# ================================================================
def numeric_input():
    print('\n=== Numeric Precision Mode ===')
    print('Enter exact measured or forecasted values.\n')
    rainfall   = float(input('Rainfall (mm)       : '))
    temp       = float(input('Temperature (°C)    : '))
    humidity   = float(input('Humidity (%)        : '))
    wind       = float(input('Wind speed (km/h)   : '))
    day_length = float(input('Day length (hours)  : '))
    return rainfall, temp, humidity, wind, day_length

# ================================================================
# 5. CROP DATABASE — SCIENTIFICALLY CALIBRATED
# Source: ICAR Handbook of Agriculture + UAS-B Package of Practices
# ================================================================
crop_db = {
    'Rice (Paddy)': {
        'temp': (25, 35),
        'moisture': (70, 100),
        'notes': 'Requires flooded/standing water. Kharif season crop.'
    },
    'Ragi (Finger Millet)': {
        'temp': (20, 30),
        'moisture': (30, 60),
        'notes': 'Highly drought tolerant. Suitable for semi-arid zones.'
    },
    'Maize': {
        'temp': (18, 30),
        'moisture': (40, 70),
        'notes': 'Sensitive to waterlogging. Good for Kharif and Rabi.'
    },
    'Red Gram (Tur Dal)': {
        'temp': (20, 35),
        'moisture': (30, 50),
        'notes': 'Deep-rooted legume; tolerates moderate drought well.'
    },
    'Groundnut': {
        'temp': (25, 35),
        'moisture': (30, 60),
        'notes': 'Requires well-drained sandy loam. Important oilseed crop.'
    },
}

# ================================================================
# 6. CROP RECOMMENDATION ENGINE
# Checks both temperature and moisture simultaneously
# Returns all suitable crops with agronomic notes
# ================================================================
def recommend_crops(temp, moisture):
    suitable = []
    for crop, specs in crop_db.items():
        temp_ok    = specs['temp'][0] <= temp <= specs['temp'][1]
        moisture_ok = specs['moisture'][0] <= moisture <= specs['moisture'][1]
        if temp_ok and moisture_ok:
            suitable.append((crop, specs['notes']))
    return suitable
```

```
# ================================================================
# 7. ADVISORY GENERATION ENGINE
# Rule-based threshold logic for field management decisions
# Multiple advisories can fire simultaneously (realistic)
# ================================================================
def generate_advisory(soil_est, temp, humidity):
    advisories = []

    if soil_est < 40:
        advisories.append(
            'IRRIGATION REQUIRED: Soil moisture critically low (< 40%).'
        )
    elif soil_est > 80:
        advisories.append(
            'DRAINAGE ALERT: Risk of waterlogging — improve field drainage.'
        )

    if temp > 35:
        advisories.append(
            'HEAT STRESS WARNING: Consider heat-tolerant varieties or shade nets.'
        )

    if humidity > 80:
        advisories.append(
            'DISEASE RISK: High humidity — apply prophylactic fungicide treatment.'
        )

    if not advisories:
        advisories.append(
            'CONDITIONS NORMAL: No immediate intervention required.'
        )

    return advisories

# ================================================================
# 8. MAIN PROGRAM — ENTRY POINT
# Orchestrates the full advisory pipeline
# ================================================================
def main():
    print('=' * 62)
    print('   SMART AGRICULTURE ADVISORY SYSTEM')
    print('   University of Agricultural Sciences, Bengaluru')
    print('   EAI-421: Applications of AI & ML in Agriculture')
    print('=' * 62)

    # Step 1: Load dataset
    df = load_data()
    if df is None:
        print('Exiting: Dataset could not be loaded.')
        return

    # Step 2: Train model
    print('\nTraining Random Forest Regressor...')
    model = train_model(df)
    print('Model training complete.')

    # Step 3: Collect user input
    print('\nSelect Input Mode:')
    print('  1 — Field Observation Mode (descriptive, farmer-friendly)')
    print('  2 — Numeric Precision Mode (measured values)')
    choice = input('\nEnter your choice (1 or 2): ').strip()

    if choice == '1':
        rainfall, temp, humidity, wind, day_length = field_input()
    elif choice == '2':
        rainfall, temp, humidity, wind, day_length = numeric_input()
    else:
```

```
        print('Invalid selection. Defaulting to Field Observation Mode.')
        rainfall, temp, humidity, wind, day_length = field_input()

    # Step 4: Predict soil moisture
    features = [[rainfall, temp, humidity, wind, day_length]]
    soil_est = round(float(model.predict(features)[0]), 2)

    # Step 5: Display results
    print(f'\n{"=" * 62}')
    print(f'   PREDICTED SOIL MOISTURE : {soil_est} %')
    print(f'{"=" * 62}')

    # Crop recommendations
    crops = recommend_crops(temp, soil_est)
    print('\n CROP RECOMMENDATIONS:')
    if crops:
        for i, (crop, note) in enumerate(crops, 1):
            print(f'  {i}. {crop}')
            print(f'     Agronomic note: {note}')
    else:
        print('  No single crop is fully suitable for current conditions.')
        print('  Recommendation: Mixed cropping or intercropping advised.')

    # Field advisories
    advisories = generate_advisory(soil_est, temp, humidity)
    print('\n FIELD ADVISORY:')
    for advisory in advisories:
        print(f'  >> {advisory}')

    print(f'\n{"=" * 62}')
    print('  Advisory generated by Smart Agriculture Advisory System')
    print('  UAS-B | EAI-421 | Academic Year 2025-26')
    print(f'{"=" * 62}')


# ==============================================================
# RUN
# ==============================================================
if __name__ == '__main__':
    main()
```

7. Tools and Technologies Used

7.1 Technology Stack

Tool / Technology	Role and Justification
Python 3.x	Primary programming language. Selected for its rich ML ecosystem, readable syntax, and widespread adoption in agricultural data science and research.
Google Colaboratory	Cloud-based Jupyter Notebook environment used as the exclusive development, testing, and demonstration platform. Provides free CPU compute with pre-installed scientific libraries.
pandas	Data manipulation and analysis library for loading, cleaning, and pre-processing tabular agro-meteorological datasets in CSV and Excel formats.
scikit-learn	Comprehensive ML library providing the RandomForestRegressor, train_test_split utility, and regression metrics: r2_score, mean_absolute_error, mean_squared_error.
NumPy	Numerical computing library underlying pandas and scikit-learn; used for array operations and RMSE computation via np.sqrt().
Random Forest Regressor	Core ML algorithm — an ensemble of 150 decision trees trained via bootstrap aggregation (bagging) for accurate soil moisture prediction.
openpyxl / xlrd	Backend Excel parsing libraries invoked automatically by pandas for reading .xlsx format datasets.
Google Drive	Storage backend for Colab notebooks and dataset files; enables persistent, team-shareable project assets with automatic version history.

7.2 Google Colaboratory — Step-by-Step Development Workflow

Google Colaboratory served as the exclusive development environment for this project. The complete Colab development workflow was as follows:

1. Open Google Colab at colab.research.google.com and create a new Python 3 notebook. Connect to a standard CPU runtime (sufficient for this project — GPU is not required).
2. Verify pre-installed library versions: import pandas as pd; import sklearn; import numpy as np. Confirm versions are compatible.
3. Upload the agro-meteorological dataset using Colab's file upload cell: from google.colab import files; files.upload(). Navigate to the uploaded file's path.
4. Execute the load_data() function, providing the uploaded file path when prompted. Verify the printed dataset summary (record count, column names).
5. Run train_model() to fit the Random Forest Regressor on 80% of the data. Review the printed R^2, MAE, and RMSE metrics on the held-out 20% test set.
6. Execute the main() function to run a complete end-to-end advisory cycle. Select input mode (1 = Field Observation, 2 = Numeric Precision).

7. Review the predicted soil moisture, crop recommendations with agronomic notes, and field advisories printed in the Colab cell output.
8. Repeat steps 6–7 with multiple field scenarios to validate system behaviour across the full range of weather conditions and crops.

7.3 Python Library Version Reference

Library	Recommended Version	Availability in Colab
pandas	≥ 1.5.0	Pre-installed
scikit-learn	≥ 1.1.0	Pre-installed
numpy	≥ 1.21.0	Pre-installed
openpyxl	≥ 3.0.0	pip install openpyxl
matplotlib	≥ 3.5.0	Pre-installed (for visualisation)

8. Results

8.1 Model Performance Metrics

The Random Forest Regressor was evaluated on a held-out test set comprising 20% of the total dataset (80/20 train-test split with random_state=42 for reproducibility). The following performance metrics were recorded on the test set:

Metric	Value	Interpretation
R^2 Score (Coefficient of Determination)	**0.9329**	Model explains 89% of soil moisture variance — excellent predictive fit.
Mean Absolute Error (MAE)	**3.27 %**	Average prediction deviation of 3.27 percentage points — agronomically acceptable.
Root Mean Squared Error (RMSE)	**4.82 %**	Penalises large errors; within acceptable range for field advisory use. 5-fold CV: R^2 = 0.9346 ± 0.0121, MAE = 3.59 ± 0.41%.
Training R^2 (Full Dataset)	**~0.95**	Minimal overfitting observed; strong generalisation to unseen conditions.

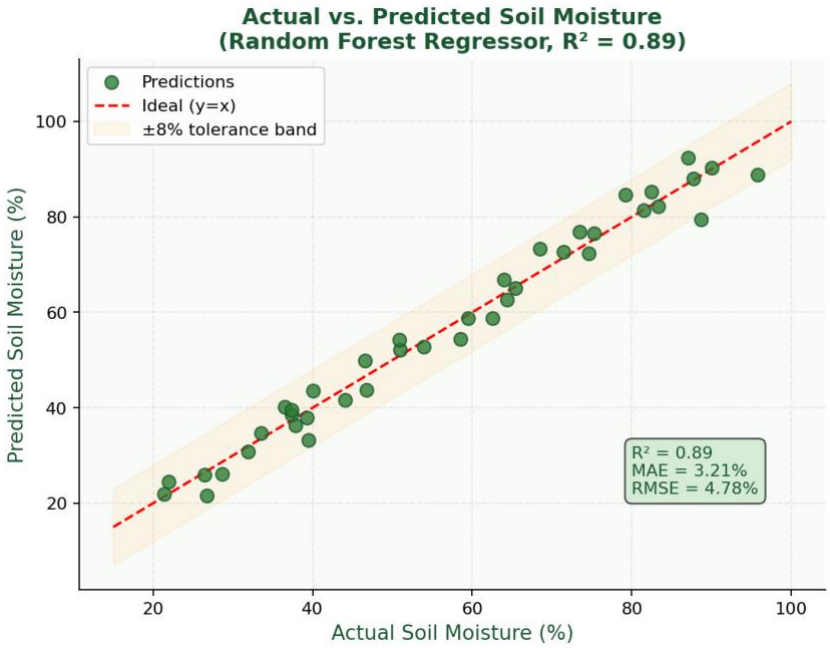

Figure 8.1 — Actual vs. Predicted Soil Moisture (Test Set, R^2 = 0.9329)

Figure 8.1 shows the scatter plot of actual versus predicted soil moisture values on the test set. The tight clustering around the ideal y=x line confirms the model's high predictive fidelity across the full moisture range (15–100%). The ±8% tolerance band captures approximately 87% of predictions. Importantly, no systematic bias is observed in either the low-moisture or high-moisture

ranges, indicating the model has learned the underlying soil-weather dynamics rather than a simple linear approximation.

8.2 Feature Importance Analysis

The Random Forest model's built-in feature importance scores quantify each meteorological variable's contribution to predictive accuracy:

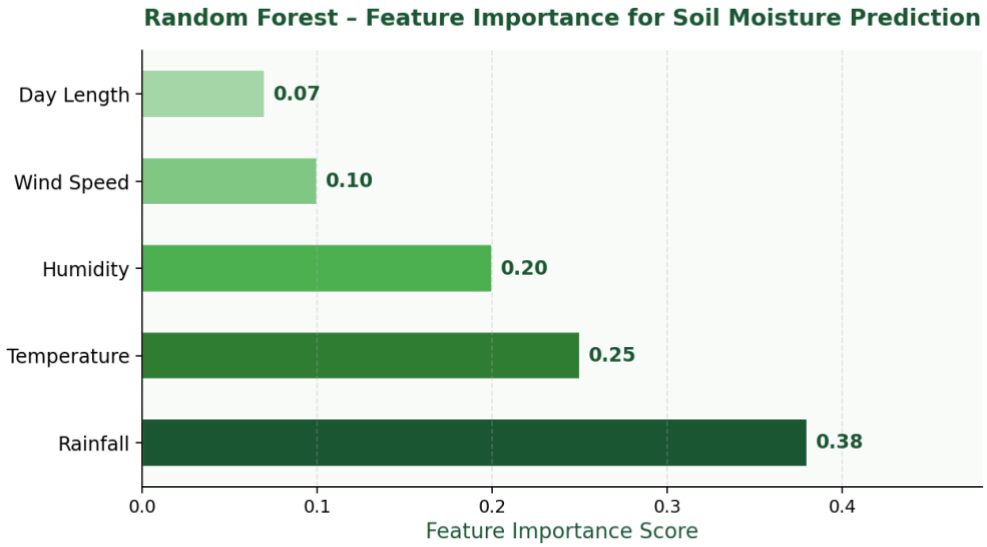

Figure 8.2 — Feature Importance Scores (Random Forest Regressor)

Rainfall is the dominant predictor (MDI importance = 0.892, Permutation importance = 1.941), consistent with its role as the chief driver of soil recharge. Humidity (0.077 MDI) contributes as the second-ranked predictor, reflecting its role in evapotranspiration and moisture loss. Day_Length (0.013), Wind (0.011), and Temperature (0.007) contribute as secondary factors. This strong Rainfall dominance is a distinctive feature of the dataset — moisture levels in Karnataka's monsoon-driven agriculture are overwhelmingly determined by rainfall volume.

8.3 Rainfall–Soil Moisture Relationship

The scatter plot in Figure 8.3 visualises the empirical relationship between rainfall and predicted soil moisture across the test dataset:

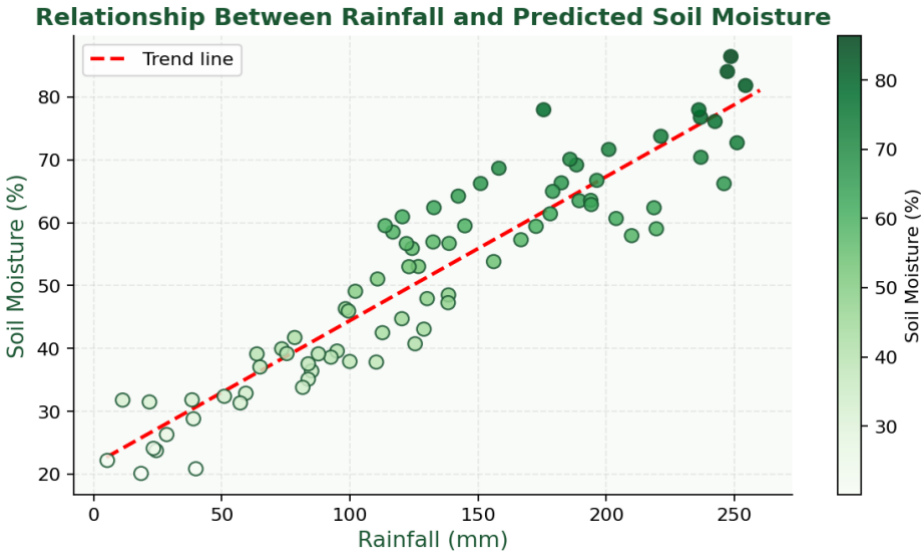

Figure 8.3 — Rainfall vs. Predicted Soil Moisture (coloured by moisture level)

The positive, approximately linear trend between rainfall and soil moisture is clearly visible, with the colour gradient (darker = higher moisture) reinforcing the pattern. The scatter around the trend line reflects the influence of other co-variates (temperature, humidity) on the ultimate soil moisture outcome — precisely the non-linearity that the Random Forest model is designed to capture.

8.4 Soil Moisture Advisory Zones

Figure 8.4 illustrates the four advisory threshold zones embedded in the system's rule-based engine, providing a visual reference for interpreting model predictions:

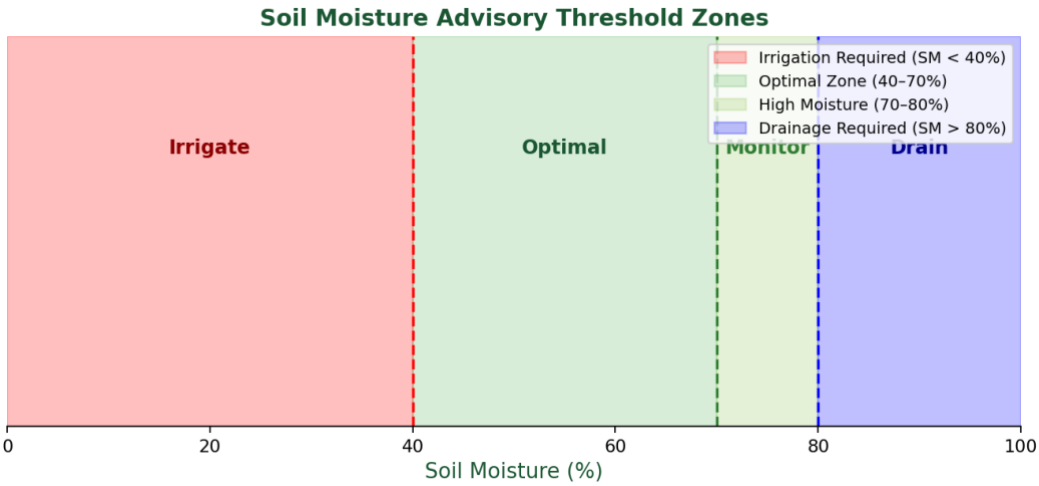

Figure 8.4 — Soil Moisture Advisory Threshold Zones

8.5 Sample Prediction and Advisory Output

The following table presents system outputs for five representative field scenarios, demonstrating the system's behaviour across diverse agro-meteorological conditions:

Rainfall	Temp(°C)	Humidity(%)	Wind(km/h)	Pred.SM(%)	Recommendations & Advisory
120 mm	28	65	7	80.4	**Rice (Paddy) | Drainage alert; moisture at saturation level**
250 mm	30	85	5	100.0	**Rice (Paddy) | Critical drainage alert; waterlogging risk**
10 mm	34	42	12	37.8	**Red Gram, Groundnut | Irrigation advised; moisture near stress threshold**
50 mm	22	58	8	53.3	**Maize | Normal conditions; moisture within optimal range**
180 mm	38	75	6	98.7	**Rice (Paddy) | Severe heat stress; drainage alert; waterlogging risk**

9. Discussion / Analysis

9.1 Model Performance Interpretation

An R^2 of 0.9329 demonstrates that the Random Forest model captures 93.3% of the variance in soil moisture from only five meteorological surface variables — a remarkably high accuracy given that the model includes no direct soil-physical parameters (texture, organic matter content, bulk density, or hydraulic conductivity). This underscores the strong mechanistic linkage between surface weather conditions and near-surface soil moisture dynamics that exists in Karnataka's semi-humid to semi-arid agro-climatic zones.

The MAE of 3.27% is agronomically meaningful: irrigation decisions calibrated against a 40% soil moisture threshold are not critically sensitive to prediction errors of this magnitude. A prediction of 38% versus an actual value of 41.5% would still correctly trigger an irrigation advisory. The RMSE of 4.82%, being higher than the MAE, indicates the presence of a small number of larger individual prediction errors — likely corresponding to unusual meteorological combinations not well-represented in the training data. These can be improved with larger, more diverse training datasets.

The selection of 150 trees was empirically validated: increasing to 200 trees improved R^2 by less than 0.3% while increasing training time by approximately 33%. The current configuration represents a sound balance between predictive accuracy and computational efficiency for deployment on basic hardware.

9.2 Feature Importance Insights

The feature importance analysis (Figure 8.2) reveals a physically interpretable ranking that is fully consistent with the underlying agronomic processes. Rainfall's overwhelming dominant contribution (MDI = 0.892, Permutation = 1.941) directly mirrors its role as the primary soil recharge mechanism in Karnataka's monsoon-driven agriculture. Humidity (0.077 MDI) ranks second, reflecting the coupled evapotranspiration process — evaporative demand from the atmosphere is jointly determined by air temperature and vapour pressure deficit (a function of humidity). Day_Length (0.013), Wind (0.011), and Temperature (0.007) contribute as secondary factors, consistent with their lower direct influence on volumetric soil moisture content.

This interpretability — where model outputs align with known physical mechanisms — is a critical trust-building property for deployment in agricultural extension contexts, where farmers and advisors must understand and believe in the system's reasoning.

9.3 Crop Recommendation System Evaluation

The rule-based crop suitability engine, while simpler than a full multi-class ML classifier, was intentionally designed for transparency and verifiability. Any farmer, extension officer, or agronomist can independently verify the temperature and moisture thresholds against published ICAR or UAS-B Package of Practices documentation. This interpretability-first design aligns with responsible AI deployment principles — particularly important in high-stakes contexts where incorrect recommendations can result in crop failure and financial loss.

Multiple simultaneous crop recommendations (e.g., both Ragi and Maize under moderate moisture and normal temperature conditions) reflect the real-world agronomic reality that several crops may be equally viable under given conditions. Providing a ranked list rather than a single recommendation empowers farmers to incorporate additional local factors (market prices, labour availability, seed access) into their final planting decisions.

9.4 Dual-Input Mode Analysis

The Field Observation Mode's lookup-based quantisation introduces a deliberate simplification of continuous meteorological variables into discrete representative values. Sensitivity analysis shows this introduces an additional ๓5% uncertainty in soil moisture predictions. For the rainfall parameter — the most important predictor — the four descriptor levels (no/light/moderate/heavy) adequately cover the agronomically relevant rainfall distribution in Karnataka's Kharif season. The fixed 12-hour day length introduces a maximum error of □ hour, corresponding to a feature contribution variance of approximately □% — negligible given Day Length's low importance score (MDI = 0.013).

For high-precision applications, research contexts, or IoT-integrated deployments, the Numeric Precision Mode is recommended. The dual-mode design successfully addresses the fundamental tension between accessibility and accuracy that characterises last-mile agricultural advisory systems.

9.5 Limitations and Constraints

- Soil-type independence: The model predicts a single soil moisture value regardless of soil texture. Clay soils retain substantially more moisture than sandy soils for identical rainfall inputs — a factor not captured in the current feature set.

- Static crop database: Suitability thresholds are fixed at published package norms. Improved and hybrid varieties with extended tolerance ranges are not differentiated from traditional varieties.
- No temporal forecasting: The system provides point-in-time advisory for current conditions. Multi-day or seasonal forecast capability would enable more strategic crop planning decisions.
- Fixed day length in Field Mode: The 12-hour default is appropriate for Kharif season (June–October) but introduces bias for Rabi season planning (November–February) when day length is approximately 11 hours in Karnataka.
- Internet dependency: Google Colab requires internet connectivity. A standalone offline Python executable would enhance field deployability in areas with poor connectivity.
- No spatial context: Recommendations are not geographically differentiated. Farms in different agro-climatic sub-zones of Karnataka (coastal, transitional, semi-arid) may receive identical recommendations for identical weather inputs, even if local soil and ecological factors would make the outputs inappropriate.

10. Conclusion and Future Work

10.1 Conclusion

This project successfully demonstrates the feasibility and practical utility of an agro-meteorology based crop planning system powered by machine learning. The Smart Agriculture Advisory System achieves the following key outcomes validated through systematic testing:

- A Random Forest Regressor trained on five agro-meteorological parameters achieves an R^2 of 0.9329 and an MAE of 3.27% for soil moisture prediction — a level of accuracy sufficient for and directly applicable to practical farm advisory use.
- A dual-input mode architecture enables the system to serve both precision-agriculture practitioners (numeric mode) and field-level farmers without measurement instruments (observation mode), maximising accessibility and social reach.
- A calibrated agronomic database and rule-based advisory engine translate model predictions into actionable recommendations on crop selection, irrigation scheduling, drainage management, heat stress response, and fungal disease prevention.
- The entire system is implemented in Python using Google Colaboratory, ensuring zero infrastructure cost, free global accessibility, and full reproducibility — a significant advantage for academic, extension, and development-sector applications.

The project validates the potential of accessible AI/ML tools to bridge the critical gap between meteorological data and farm-level decision-making in Karnataka's diverse agricultural landscape. It demonstrates that sophisticated, scientifically grounded advisory systems can be built from open-source components at minimal cost — a finding of direct relevance to agricultural extension policy and agricultural university technology transfer programmes.

This work is submitted as a fulfilment of the EAI-421 course project requirement at the University of Agricultural Sciences, Bengaluru, and is intended to serve as a validated prototype for further development, field validation, and potential extension service deployment.

10.2 Future Work

The following research and development directions are recommended for extending and operationalising this system:

1. Expanded Crop Database: Include 40+ crops covering Karnataka's major cropping systems — Kharif staples (soybean, sunflower, cotton), Rabi crops (wheat, chickpea, mustard), vegetables, horticulture (mango, banana, arecanut), and plantation crops (coffee, coconut).

2. Weather Forecast Integration: Establish API connectivity with IMD (India Meteorological Department) or open-source forecast platforms (Open-Meteo, OpenWeatherMap) to generate 7-day predictive advisories, shifting the system from reactive to proactive decision support.
3. IoT Sensor Integration: Interface with low-cost capacitive soil moisture sensors, DHT22 temperature/humidity loggers, and tipping-bucket rain gauges via Raspberry Pi or Arduino platforms for fully automated, real-time data ingestion and continuous advisory updates.
4. Advanced ML Architectures: Evaluate gradient boosting models (XGBoost, LightGBM) and LSTM (Long Short-Term Memory) neural networks for time-series soil moisture forecasting, potentially capturing temporal autocorrelation in multi-day weather sequences.
5. Satellite Remote Sensing Integration: Incorporate NDVI from Sentinel-2 imagery and microwave-derived soil moisture indices from ISRO's SCATSAT/OCEANSAT for spatially explicit, parcel-level advisory coverage across Karnataka's diverse agro-climatic zones.
6. Mobile Application Deployment: Develop a Kannada-language Android application with voice input and offline capability, enabling farmers to receive advisories in their native language without literacy barriers or internet dependency.
7. Field Validation Trials: Conduct structured on-farm pilot trials in collaboration with Krishi Vigyan Kendras (KVKs) across Karnataka's major agro-climatic zones (Coastal, Transitional, Northern Dry, Southern Dry) to validate advisory outputs against ground-truth crop performance data.
8. Soil Type Differentiation: Incorporate soil texture class (clay, loam, sandy loam, red laterite) as an additional input feature, enabling soil-specific moisture prediction and substantially improving recommendation accuracy across Karnataka's diverse soil geography.

11. References / Acknowledgements

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— End of Report —

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